

Review of Experimental Data and Evaluation of Equations of State for Modeling Formation of Solid CO₂ in CCS and Natural Gas Applications

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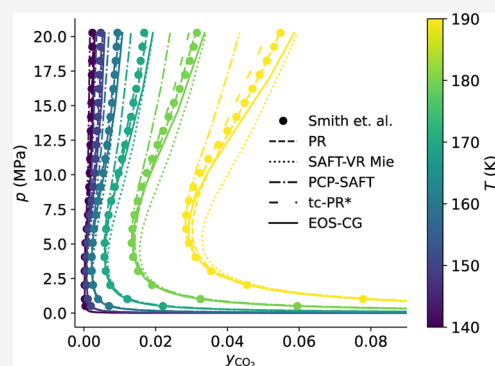
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ABSTRACT: Solid CO₂ is a potential safety hazard in many applications. Freeze-out can be predicted by combining the equations of state (EoS) for the fluid and solid phases. We have reviewed available equilibrium data for mixtures relevant for carbon capture, transport, and storage (CCS) and natural gas applications and performed a comprehensive assessment of model combinations. The most significant difference in model accuracy came from the choice of fluid EoS. For CCS applications, the best agreement with experimental data was achieved by using the Peng–Robinson EoS, yielding a Mean Absolute Percentage Deviation (MAPD) of 1.4%. For natural gas applications, the most updated multiparameter EoS was the most accurate choice, yielding a MAPD below 1%. Provided that an accurate EoS for the solid phase is available, including solid data in the regression of the fluid EoS represents an avenue that can be exploited to achieve more accurate solid-phase predictions in the future.



1. INTRODUCTION

To mitigate global warming, a growing number of carbon capture, transport, and storage (CCS) projects are currently being planned and executed.¹ In these projects, CO₂ will be captured from different emission sources such as coal and gas power plants, cement, chemical, and steel production facilities, and steam reforming plants for hydrogen production.² This will result in a wide spectrum of potential components in the CO₂ mixture, where a comprehensive list and expected composition ranges were presented by Li et al.³

An important safety assessment in CCS is to evaluate the risk of forming solid CO₂, also called dry ice. If the temperature becomes sufficiently low, solid CO₂ will form either from a gas phase at lower pressures or from a liquid phase at higher pressures. Solid CO₂ can also precipitate in Liquefied Natural Gas (LNG) applications,⁴ cryogenic biogas upgrading,⁵ air-separation units,⁶ and several other examples.⁶ The exact conditions where this occurs depend on the type and amount of components present in the mixture.

Formation of solid CO₂ can cause several problems. If solid CO₂ forms locally, it can plug essential equipment, such as vent lines or safety relief systems. If a solid CO₂ plug detaches from the wall and gains momentum, it can damage equipment and pose a serious risk to personnel.⁷ The formation of solid CO₂ can also cause pressure surges or ruptures in the pipeline.⁸

Many scenarios can trigger the formation of dry ice. During a depressurization event due to a pipeline shutdown, rupture, or safety valve activation, the Joule–Thomson effect will lead to rapid cool down.⁸ Depressurization or loss of temperature

control in storage tanks can also lead to very low temperatures.⁹ If control is lost over the wellhead valves during injection into a reservoir in case of a blow-out^{10,11} or if significant cooling happens during startup or shut-in of a well, dry ice may form and cause overpressure, loss of injectivity, and wellhead damage.

A promising technology to capture CO₂ from sources such as synthesis gas from coal gasification, called low-temperature CO₂ capture, has shown to be competitive compared to baseline solvent-based capture technologies.¹² To further improve and optimize this technology for various applications, it is important to have accurate control of the limits of dry ice formation.¹³

Two very accurate equations of state (EoS) for solid CO₂ have been published. Trusler published a Helmholtz energy-based EoS,^{14,15} while Jäger and Span presented a Gibbs free energy function.¹⁶ Siah et al.¹⁷ recently developed a new Gibbs energy-based EoS, but in their comparisons with the aforementioned models, the equation of Trusler generally performed best. Siah et al. also performed predictions for mixtures with methane and nitrogen. When combined with an EoS for the fluid mixture, the solid-phase EoS can be used to predict when dry ice forms in mixtures.

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Several studies have compared predictions from this approach to experimental data. In 2016, Nikolaidis et al. used cubic equations of state and PC-SAFT to describe the fluid phase, and the EoS by Jäger and Span to describe the solid phase,¹⁶ and compared the predictions to experimental results for CO₂ with several other components such as nitrogen, hydrogen, and methane.¹⁸ The authors found a reasonable agreement between theory and experiments but observed larger deviations at higher pressures. Lovseth et al.¹⁹ analyzed the CO₂–argon system utilizing several EoS, including EoS-CG,^{20,21} GERG-2008,²² and cubic EoS combined with the solid EoS by Jäger and Span, and obtained excellent agreement with experiments. In 2019, Tang et al. used GERG-2004²³ to describe the fluid phase and the EoS by Jäger and Span to describe the solid phase. Predictions were compared to experimental data for the CO₂–CH₄ system.²⁴ The model prediction of the solid–liquid–vapor locus and liquid-phase composition along the three-phase locus at higher temperature ranges (between 205 K and a CO₂ triple-point temperature of 216.5 K) was not satisfactory.

Yang et al.²⁵ performed several Solid Fluid Equilibrium (SFE) predictions for CO₂ mixtures with focus on LNG. They tested several cubic equations of state, combined with a fugacity model for the solid phase. The Peng–Robinson (PR) EoS performed best with a standard uncertainty below 2 K.

Focusing on natural gas purification and biogas upgrading, De Guido and Spatolisano presented in 2021 a comparison for CO₂–methane with components such as oxygen and nitrogen using the Peng–Robinson cubic EoS for the fluid phase and a six-parameter correlation for the solid phase. They deemed the performance to be satisfactory.²⁶ Ababneh and Al-Muhtaseb used in 2021 a similar approach for the CO₂–CH₄–H₂S system.²⁷ Riva et al.⁵ assessed the CO₂–CH₄ system utilizing two models: the PR EoS coupled with a simplified solid fugacity model and an analytical EoS by Yokozeki²⁸ capable of simultaneously representing solid, vapor, and liquid phases. Both had similar performances and were deemed satisfactory.

Campestrini et al.²⁹ performed a review of the CO₂/H₂S system, analyzing the solidification of both components. They utilized both cubic and GERG-2008 EoS for the fluid phase and a fugacity model for the solid phases. Both fluid models gave similar accuracy in representing the global phase equilibrium behavior of the mixture.

Siah et al. presented in 2025 a new algorithm to perform phase-equilibrium calculations with several solid phases.³⁰ In this work, they used the GERG-2008 EoS for the fluid phase and the solid EoS by Jäger and Span, demonstrating good agreement with experimental data for the CO₂–CH₄ system. The most accurate and updated reference EoS for CO₂–mixtures is a multiparameter EoS called EoS-CG^{20,21} with updated mixture departure functions.^{31,32}

The first goal of this article is to review available experimental data on SFE with solid CO₂ for mixtures relevant for CCS and natural gas applications. Mixtures with water have been excluded since ice and hydrates fall beyond the scope of this work. Next, we critically evaluate the accuracy of the most accurate model combinations that are available. This makes it possible to establish a benchmark choice of models to estimate when dry ice forms from mixtures and to estimate the uncertainty of this model combination. The most accurate model combination can be used to maintain the design and operation of infrastructure within safe boundaries and avoid or handle the formation of solid CO₂. A second goal of the article is to provide guidelines

for how different types of EoS can be extended or modified to improve their accuracy in the prediction of SFE.

2. EQUATIONS OF STATE

Equations of state for the solid phase and the fluid phase are needed to predict SFE. The different EoS that have been considered in this work are presented in Sections 2.1–2.2.

2.1. EoS for the Solid Phase. Two EoS for solid CO₂ have been assessed in this work. The Helmholtz-energy-based EoS by Trusler^{14,15} is founded on the Quasi-Harmonic approximation, with additional contributions from anharmonicities and internal vibrations. The molar Helmholtz energy is

$$A_m(T, \nu) = A_0(\nu) + A_D(T, \nu) + \sum_{i=1}^5 a_i A_i(T, \nu) + A_A(T, \nu) \quad (1)$$

where T is the temperature, ν is the molar volume, A_0 is the “cold-curve” energy, A_D and A_i refer to vibrational energies, and A_A accounts for anharmonicity. The model was shown to be valid up to temperatures and pressures of 800 K and 12 GPa.

The EoS by Jäger and Span¹⁶ is formulated in terms of the molar Gibbs energy:

$$G_m(p, T) = h_0 - Ts_0 + \int_{T_0}^T c_p(T', p_0) dT' - T \int_{T_0}^T \frac{c_p(T', p_0)}{T'} dT' + \int_{p_0}^p v(p, T) dp \quad (2)$$

where c_p is the isobaric heat capacity along an isobar p_0 , ν is a thermal equation of state explicit in the molar volume, and h_0 and s_0 are integration constants. The full expression for the EoS is provided in eq 8 in ref 16. The EoS is valid for pressures from 0 MPa to 0.5 GPa and temperatures from 80 to 300 K. Other fugacity-based models for solid CO₂ have been presented in the literature,²⁵ but typically with a lower accuracy than the EoS by Trusler^{14,15} and Jäger and Span.¹⁶

2.2. EoS for the Fluid Phase. We have included several fluid-phase EoS in the discussion for comparison. Since they are a popular choice in process simulations and computational fluid dynamics studies, we have included for comparison the cubic EoS Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR).^{33,34} We have also included an improved version of PR called translation-consistent PR (tc-PR).³⁵ Three EoS founded in statistical associating fluid theory (SAFT) have been assessed: PC-SAFT,³⁶ PCP-SAFT,³⁷ and SAFT-VR-Mie.³⁸ The most recent multiparameter EoS reference for natural gas called GERG-2008²² and the most accurate and updated multiparameter EoS called EoS-CG^{20,21} with updated mixture departure functions^{31,32} have also been included. To include SO₂ with GERG-2008, we extend the model using the multiparameter EoS of Gao et al.³⁹ and the binary interaction models of EoS-CG.

We believe that this collection of fluid phase EoS is sufficiently broad to facilitate an in-depth discussion of how the fluid EoS influences the accuracy of the predictions.

3. METHODOLOGY

3.1. Combining the Fluid and Solid EoS. Since the reference states differ for the fluid and solid EoS, the reference state of the solid-phase EoS has been adjusted to match the

Gibbs energy and the melting entropy of the fluid EoS for pure CO₂ at the triple point. The triple-point temperature was set to the value of $T_{tr} = 216.592$ K, and the triple-point pressure was set corresponding to the saturation pressure from the fluid EoS. The molar Gibbs energy, g_{solid} , of the solid was adjusted to g_{solid}^* using

$$g_{solid}^*(T, p) = g_{solid}(T, p) + \Delta g_{tr} - (T - T_{tr})\Delta s_{tr} \quad (3)$$

Here, the constant molar Gibbs energy adjustment is

$$\Delta g_{tr} = g_f(T_{tr}, p_{tr}) - g_{solid}(T_{tr}, p_{tr}) \quad (4)$$

where g_f is the liquid molar Gibbs energy calculated from the fluid EoS. The adjustment of the melting entropy is

$$\Delta s_{tr} = s_f(T_{tr}, p_{tr}) - s_{solid}(T_{tr}, p_{tr}) - \Delta s_{melt, tr} \quad (5)$$

where $\Delta s_{melt, tr}$ is the entropy of melting at the triple point, which was chosen to be $\Delta h_{melt, tr}/T_{tr}$, where $\Delta h_{melt, tr} = 8875$ J/mol. Note that g_{solid} and s_{solid} are obtained from the solid EoS of the original works independently of the fluid EoS.

The ideal gas model will also affect the predictions and should therefore be consistent when comparing different model combinations. For all fluid EoS, we have used the ideal gas models with the highest accuracy, i.e., those available in the most updated multiparameter EoS, EoS-CG.

3.2. The α -Function of tc-PR below the Triple Point. In the pioneering work presenting tc-PR, the α -function was regressed by considering thermodynamic properties above the triple point (see eq 4 in ref 35). The expression used for the α -function is the three-parameter T_w function,³⁵

$$\alpha(T) = T_r^{N(M-1)} \exp\{[L(1 - T_r^{MN})]\} \quad (6)$$

where $T_r = T/T_{crit}$ is the reduced temperature and L , M , and N are parameters. In this work, we shall discuss the effect of also including experimental data from the sublimation curve below the triple point. The modified objective function when determining a new α -function is then:

$$\begin{aligned} OBJ = & \frac{w_\alpha}{N_\alpha} \sum_{i=1}^{N_\alpha} \left[\frac{\alpha_{tc-PR}(T_i) - \alpha_{tc-PR^*}(T_i)}{\alpha_{tc-PR}(T_i)} \right]^2 \\ & + \frac{w_p}{N_p} \sum_{i=1}^{N_p} \left[\frac{p_{Sub, Aux}(T_i) - p_{Sub, tc-PR^*}(T_i)}{p_{Sub, Aux}(T_i)} \right]^2 \\ & + \frac{w_h}{N_h} \sum_{i=1}^{N_h} \left[\frac{\Delta h_{Sub, EoS-CG}(T_i) - \Delta h_{Sub, tc-PR^*}(T_i)}{\Delta h_{Sub, EoS-CG}(T_i)} \right]^2 \end{aligned} \quad (7)$$

where N_i is the number of experiments used for the property in question and subscript i refers to the property in question, which are α , sublimation pressure, and sublimation enthalpy. The variables w_i are weighting factors, where $w_\alpha = 1$, $w_p = 40$, and $w_h = 800$ were used to ensure a similar performance for the vapor–liquid curve and properties in the supercritical domain as the original tc-PR, but with improved performance for the sublimation curve. In determining the modified α -function, we used the solid EoS by Trusler to determine sublimation pressures (p_{Sub}) and enthalpies (h_{Sub}). Furthermore, $p_{Sub, Aux}$ is an auxiliary equation for sublimation pressure representing the experimental data nearly within their uncertainty,¹⁴ and EoS-CG was used to determine the sublimation enthalpy. For α_{tc-PR} , an equidistant grid ranging from the triple point to 300 K was generated to ensure reliable fluid-phase predictions above the

triple point. For the sublimation properties, a grid ranging between 130 K and the triple point was generated. The revised parameters shown in Table 1 also pass the consistency criteria of

Table 1. Revised Parameters for α for the tc-PR EoS, Where c is the Volume Shift

Component	L	M	N	c (cm ³ /mol)
CO ₂	1.1241	0.9999	0.6074	−1.5738

tc-PR⁴⁰ but result in a slightly lower accuracy for the properties above the triple point; see the Supporting Information (SI) for comparisons. In addition to modifying the α -function, we have modified the volume shift by matching the density at the triple point.

3.3. Assessing the Model Combinations. The performance of the model combinations will be assessed in terms of the Mean Absolute Percentage Deviation (MAPD) of the predicted phase-equilibrium temperature, which is defined as

$$MAPD = \frac{1}{N_{Points}} \sum_{i=1}^{N_{Points}} \left| \frac{T_{i, EXP} - T_i}{T_{i, EXP}} \right| \cdot 100\% \quad (8)$$

where T_i is the temperature predicted by the model combination with the measured temperature $T_{i, EXP}$ for experiment i , assuming that composition and pressure are given. In principle, other metrics for assessing the performance of the theory could also be considered, such as the MAPD of composition or pressure. We find that these metrics provide very similar information to the MAPD defined in eq 8 and have therefore not included them in the discussion in Section 4.

4. RESULTS AND DISCUSSION

After testing the two EoS for solid CO₂ presented in Section 2.1, we found them to give nearly identical results in SFE predictions. The solid EoS presented by Trusler, however, yielded slightly lower overall deviations and will be used in the remaining part of this work. The SFE measurements relevant for CCS and natural gas applications that have been presented in the last century are summarized in Table 2.^{44–51} To compare the fluid EoS on an equal basis, all relevant binary interaction parameters were regressed using the same experimental data, as discussed in the SI. The MAPD for all EoS and mixtures is summarized in Table 3, where the last lines represent the average MAPD for the fluid EoS with each experimental set (not each point) weighted equally.

4.1. Available Experimental Data. The available experimental data can be categorized into Solid Vapor Equilibria (SVE), Solid Liquid Equilibria (SLE), Solid Vapor Liquid Equilibria (SVLE), and SVLE-locus measurements. The uncertainty of the measurements is typically reported to be in the range of 1–25 kPa for the pressure and within 0.1 K for the temperature, with some notable exceptions. Few experiments have considered a high CO₂ content relevant for CCS applications, and the other components typically have the highest mole fractions in the available experimental data.

Kurata and coworkers^{45,56,67} studied the formation of solid CO₂ in mixtures with a variety of hydrocarbons. They measured SLE, SVLE, and SVLE loci.

The CO₂/N₂ system has been studied for SVE,^{54,55} SLE,⁵⁷ and SVLE loci.⁶⁸ Løvseth et al.¹⁹ measured both SLE and SVE for a CO₂/Ar mixture. The CO₂/H₂S system has also been the

Table 2. Overview of the Experimental Data for the CO₂ Mixtures^a

Mixture	SFE	Year	N _{points}	x _{CO₂} range (%)	T range (K)	p range (MPa)
CO ₂ /CH ₄	SVE ⁴⁸	2011	17	[10.8000, 54.2000]	[191.1, 210.3]	[0.2930, 4.4460]
CO ₂ /CH ₄	SVE ⁴⁷	2007	55	[1.0000, 2.9300]	[168.6, 187.7]	[0.9621, 3.0082]
CO ₂ /C ₂ H ₆ /CH ₄	SVE ⁴⁷	2007	24	[1.9500, 1.9600]	[173.6, 183.6]	[1.2677, 2.2503]
CO ₂ /N ₂ /CH ₄	SVE ⁴⁷	2007	24	[1.9400, 1.9400]	[173.9, 183.5]	[1.2437, 2.2615]
CO ₂ /CH ₄	SVE ⁴¹	1959	38	[1.0000, 20.0000]	[158.2, 210.5]	[0.1966, 4.8322]
CO ₂ /CH ₄	SVE ^{52,53}	1998	43	[2.0000, 16.0000]	[170.1, 199.8]	[0.6893, 2.7571]
CO ₂ /CH ₄	SVE ⁴⁶	1974	42	[0.1200, 10.6700]	[137.5, 198.1]	[0.1724, 2.7855]
CO ₂ /N ₂ /CH ₄	SVE ⁴⁶	1974	19	[0.9300, 0.9600]	[154.1, 172.8]	[0.1724, 2.4407]
CO ₂ /CH ₄	SVE ⁵¹	2015	64	[0.1000, 34.0700]	[153.2, 193.2]	[0.2190, 3.0380]
CO ₂ /CH ₄ /N ₂	SVE ⁵¹	2015	77	[0.1200, 34.7900]	[153.2, 193.2]	[0.2670, 2.2000]
CO ₂ /CH ₄ /C ₂ H ₆	SVE ⁵¹	2015	53	[0.6500, 33.6500]	[153.2, 193.2]	[0.2720, 2.1950]
CO ₂ /CH ₄	SVE ⁴¹	1959	66	[0.0265, 58.0000]	[132.6, 210.1]	[0.1560, 4.7896]
CO ₂ /N ₂	SVE ^{54,55}	1963	125	[0.0180, 7.7800]	[140.0, 190.0]	[0.5066, 20.2650]
CO ₂ /Ar	SVE ¹⁹	2018	4	[13.1320, 42.3570]	[213.1, 213.1]	[1.0495, 6.8449]
CO ₂ /CH ₄	SLE ⁴	2023	6	[0.0052, 0.0500]	[106.3, 122.8]	[6.9800, 10.1000]
CO ₂ /CH ₄	SLE ⁵⁶ (a)	1974	11	[0.1600, 20.5000]	[129.7, 201.3]	[5.0000, 5.0000]
CO ₂ /CH ₄	SLE ⁴⁹	2012	9	[0.0213, 2.8960]	[112.0, 169.9]	[0.0930, 2.3150]
CO ₂ /CH ₄ /N ₂	SLE ⁴⁹	2012	27	[0.0225, 2.7700]	[112.0, 169.9]	[0.1550, 3.1500]
CO ₂ /CH ₄ /C ₂ H ₆	SLE ⁴⁹	2012	27	[0.0577, 3.3100]	[120.0, 169.9]	[0.1620, 2.2280]
CO ₂ /CH ₄ /N ₂	SLE ⁵⁰	2012	32	[0.0003, 0.0900]	[83.1, 123.1]	[0.1240, 2.0230]
CO ₂ /CH ₄	SLE ⁴¹	1959	23	[0.0320, 20.0000]	[113.5, 201.3]	[1.0132, 9.7272]
CO ₂ /Ar	SLE ¹⁹	2018	3	[78.6300, 86.6210]	[213.1, 213.1]	[6.8455, 16.1441]
CO ₂ /N ₂	SLE ⁵⁷	1975	29	[0.0002, 0.0214]	[78.0, 115.0]	[3.6477, 9.1192]
CO ₂ /O ₂	SLE ⁵⁸ (a)	2002	12	[0.0004, 0.0098]	[90.77, 110.4]	[5.8900, 5.8900]
CO ₂ /O ₂	SLE ⁵⁹	1986	14	[0.00001, 0.2068]	[77.4, 140.0]	[5.8900, 9.5000]
CO ₂ /H ₂ S	SLE ⁶⁰ (b)	1959	12	[12.3473, 99.7893]	[178.4, 216.8]	[0.0214, 0.4176]
CO ₂ /H ₂ S	SLE ⁶¹	1932	9	[9.7900, 25.0100]	[177.9, 191.1]	[0.0338, 0.1018]
CO ₂ /H ₂ S	SLE ⁶²	1920	1	[25.4000, 25.4000]	[191.1, 191.1]	[0.1000, 0.1000]
CO ₂ /SO ₂	SLE ⁶²	1920	1	[33.6000, 33.6000]	[194.5, 194.5]	[0.1000, 0.1000]
CO ₂ /CH ₄ /C ₂ H ₆	SVLE ⁴⁵	1971	10	[3.3600, 64.4600]	[166.5, 208.2]	[0.9025, 3.0489]
CO ₂ /CH ₄ /C ₃ H ₈	SVLE ⁴⁵	1971	12	[1.0300, 62.0100]	[150.2, 208.2]	[0.4571, 2.3649]
CO ₂ /CH ₄ /C ₄ H ₁₀	SVLE ⁴⁵	1971	13	[10.4800, 78.1200]	[185.2, 210.2]	[1.2755, 3.3302]
CO ₂ /C ₂ H ₆ /C ₃ H ₈	SVLE ⁴⁵	1971	8	[11.8300, 63.2000]	[190.2, 208.2]	[0.1207, 0.4619]
CO ₂ /CH ₄ /C ₂ H ₆ /C ₃ H ₈	SVLE ⁴⁵	1971	20	[14.7700, 67.4100]	[190.2, 208.2]	[1.3790, 2.7579]
CO ₂ /CH ₄ /N ₂	SVLE ⁶³	2018	6	[0.0532, 0.6909]	[124.5, 145.9]	[0.5200, 2.0400]
CO ₂ /CH ₄ /N ₂ /O ₂	SVLE ⁶³	2018	6	[0.0356, 0.6948]	[125.0, 146.5]	[0.5800, 2.1100]
CO ₂ /CH ₄ /C ₂ H ₆	SVLE ⁶⁴	2022	53	[0.6195, 38.8250]	[148.2, 203.16]	[0.2200, 3.868]
CO ₂ /CH ₄	SVLE ⁶⁵	2020	7	NA	[204.0, 216.0]	[0.9640, 4.8580]
CO ₂ /CH ₄	SVLE ⁶⁶	2015	15	NA	[190.4, 210.4]	[3.9790, 4.5670]
CO ₂ /CH ₄	SVLE ⁴³	1962	30	NA	[97.5, 211.7]	[0.0283, 4.8677]
CO ₂ /CH ₄	SVLE ⁴²	1954	20	NA	[194.5, 215.7]	[0.9170, 4.8539]
CO ₂ /CH ₄	SVLE ⁴⁴	1961	5	NA	[166.4, 199.6]	[1.9351, 5.0314]
CO ₂ /C ₂ H ₆	SVLE ⁶⁷	1971	13	NA	[99.8, 214.8]	[0.0005, 0.5664]
CO ₂ /C ₃ H ₈	SVLE ⁶⁷	1971	11	NA	[141.5, 213.7]	[0.0009, 0.4489]
CO ₂ /C ₄ H ₁₀	SVLE ⁶⁷	1971	12	NA	[141.5, 210.9]	[0.0008, 0.3557]
CO ₂ /N ₂	SVLE ⁶⁸	2015	4	NA	[212.7, 214.9]	[4.8230, 13.0180]
CO ₂ /H ₂	SVLE ⁶⁸	2015	3	NA	[217.0, 217.9]	[4.3000, 13.7420]

^aReferences for which we assume the pressure to either be constant (a) or equal (b) to that of the SVLE curve are marked.

focus of at least three studies,^{60–62} where Thiel and Schulte investigated several systems, including CO₂/SO₂.⁶²

There are more solubility studies available than those reported in Table 2. Most of these studies have not reported the pressure, which makes a reliable comparison to the models difficult. They have, therefore, not been included or discussed in this work. Some exceptions are the CO₂/CH₄, CO₂/H₂S, and CO₂/O₂ systems, where the pressure can be reliably deduced from the information available in the respective articles. Some omitted references for the CO₂/CH₄, CO₂/O₂, CO₂/N₂, and CO₂/Ar systems are refs 69–75. We have also excluded one

experimental study which exhibited unreasonable disagreement with all of the model combinations.⁷⁶

It should be noted that some of the reported SLE experiments are in fact SVLE experiments as there is coexistence with gas. However, we have chosen to categorize these as SLE when there are no reported vapor measurements or if there is no clear statement in the article that there is coexistence between three phases.

4.1.1. SFE for Pure CO₂. Before assessing the theories for mixtures, it makes sense to compare their predictions for the solid formation of pure CO₂. This is especially relevant for CCS

Table 3. Comparison of MAPD Values across Different EoS Models^a

Mixture	SFE	EoS-CG	GERG	tc-PR*	PR	SRK	VR-Mie	PC	PCP
CO ₂ /CH ₄	SVE ⁴⁸	0.34	0.34	0.36	0.34	0.37	0.32	0.77	0.33
CO ₂ /CH ₄	SVE ⁴⁷	1.38	1.38	1.26	1.29	1.25	1.33	1.93	1.40
CO ₂ /C ₂ H ₆ /CH ₄	SVE ⁴⁷	1.00	1.00	0.96	0.95	0.96	0.96	1.38	0.93
CO ₂ /N ₂ /CH ₄	SVE ⁴⁷	0.95	0.95	0.92	0.91	0.92	0.93	1.38	0.93
CO ₂ /CH ₄	SVE ⁴¹	0.32	0.32	0.34	0.37	0.36	0.39	1.27	0.51
CO ₂ /CH ₄	SVE ^{52,53}	0.64	2.03	0.74	0.67	0.77	0.58	0.45	0.39
CO ₂ /CH ₄	SVE ⁴⁶	0.99	0.99	1.04	0.97	1.06	0.95	1.04	0.85
CO ₂ /N ₂ /CH ₄	SVE ⁴⁶	0.68	0.68	0.78	0.69	0.81	0.64	0.77	0.49
CO ₂ /CH ₄	SVE ⁵¹	0.48	0.48	0.48	0.45	0.48	0.43	1.26	0.48
CO ₂ /CH ₄ /N ₂	SVE ⁵¹	0.58	0.58	0.57	0.57	0.57	0.56	1.37	0.63
CO ₂ /CH ₄ /C ₂ H ₆	SVE ⁵¹	0.59	0.59	0.63	0.56	0.64	0.49	0.58	0.37
CO ₂ /CH ₄	SVE ⁴¹	0.21	0.21	0.37	0.30	0.42	0.26	0.86	0.25
CO ₂ /N ₂	SVE ^{54,55}	0.93	0.93	0.38	0.36	0.36	1.52	1.89	1.67
CO ₂ /Ar	SVE ¹⁹	0.03	0.09	0.09	0.04	0.19	0.10	0.92	0.10
CO ₂ /CH ₄	SLE ⁴	2.97	2.98	5.75	5.12	5.72	10.32	8.53	3.67
CO ₂ /CH ₄	SLE ⁵⁶ (a)	0.50	0.50	0.88	0.85	0.94	2.08	2.67	0.67
CO ₂ /CH ₄ /N ₂	SLE ⁴⁹	0.63	0.63	0.68	0.43	0.81	3.56	4.10	0.79
CO ₂ /CH ₄ /C ₂ H ₆	SLE ⁴⁹	0.36	0.36	0.93	0.65	1.02	3.75	4.01	0.50
CO ₂ /CH ₄	SLE ⁴⁹	0.45	0.45	0.81	0.59	0.94	3.79	4.01	0.90
CO ₂ /CH ₄ /N ₂	SLE ⁵⁰	1.22	1.21	1.42	1.00	1.60	4.38	4.65	1.73
CO ₂ /CH ₄	SLE ⁴¹	0.70	0.70	1.03	1.00	1.00	1.63	2.22	0.79
CO ₂ /Ar	SLE ¹⁹	0.03	0.09	0.24	0.45	0.15	0.25	0.26	0.30
CO ₂ /O ₂	SLE ⁵⁸ (a)	11.98	6.20	3.30	2.45	2.82	11.58	14.00	0.74
CO ₂ /O ₂	SLE ⁵⁹	11.57	6.02	3.06	2.33	2.57	10.90	13.53	0.64
CO ₂ /N ₂	SLE ⁵⁷	2.45	2.45	1.23	0.57	1.87	1.74	2.68	6.22
CO ₂ /H ₂ S	SLE ⁶⁰ (b)	0.60	1.56	0.51	0.52	0.54	5.28	3.10	1.99
CO ₂ /H ₂ S	SLE ⁶¹	1.95	1.98	1.26	1.32	1.40	18.38	6.65	6.79
CO ₂ /H ₂ S	SLE ⁶²	0.42	0.41	1.73	1.76	1.81	8.39	4.36	4.64
CO ₂ /SO ₂	SLE ⁶²	3.61	3.65	1.36	1.22	6.22	8.53	27.37	6.25
CO ₂ /CH ₄ /C ₂ H ₆	SVLE ⁴⁵	0.32	0.32	0.40	0.37	0.40	0.89	1.55	1.65
CO ₂ /CH ₄ /C ₃ H ₈	SVLE ⁴⁵	1.04	3.25	1.59	1.49	1.50	2.74	8.75	1.17
CO ₂ /CH ₄ /C ₄ H ₁₀	SVLE ⁴⁵	0.35	0.35	0.78	0.75	0.78	1.19	1.68	1.70
CO ₂ /C ₂ H ₆ /C ₃ H ₈	SVLE ⁴⁵	0.83	1.10	1.05	1.04	1.03	1.57	5.11	1.20
CO ₂ /CH ₄ /C ₂ H ₆ /C ₃ H ₈	SVLE ⁴⁵	0.48	0.48	0.77	0.50	0.51	0.65	3.16	0.69
CO ₂ /CH ₄ /N ₂	SVLE ⁶³	2.80	2.80	3.13	2.85	3.23	5.26	6.15	2.46
CO ₂ /CH ₄ /N ₂ /O ₂	SVLE ⁶³	3.92	3.76	5.89	3.94	4.24	6.33	7.34	3.39
CO ₂ /CH ₄ /C ₂ H ₆	SVLE ⁶⁴	0.68	0.67	0.60	0.55	0.65	2.01	2.90	2.06
MAPD	SVE	0.65	0.76	0.64	0.60	0.65	0.68	1.13	0.67
MAPD	SLE	2.63	1.95	1.61	1.35	1.96	6.30	6.81	2.44
MAPD	SVLE	1.30	1.59	1.78	1.44	1.54	2.58	4.58	1.79
MAPD		1.59	1.42	1.28	1.09	1.38	3.37	4.18	1.63

^aReferences in which we assume pressure to be either constant (a) or equal (b) to that of the SVLE curve are marked. For tc-PR*, we utilized the parameters in Table 1 for CO₂.

applications since the CO₂-mixture often contains a small amount of impurities. The auxiliary functions presented by Trusler¹⁴ have been used as a reference. They represent the experimental data for the melting curve within 2.5% and the experimental data for the sublimation curve below 1–2% in the temperature range considered.

A comparison of the different choices of fluid EoS is presented in Figure 1 for both the melting and sublimation curves. For the melting curve, Figure 1a shows that it is necessary to incorporate a volume shift in cubic EoS to achieve reasonable agreement with experimental data, as both PR and SRK exhibit deviations exceeding 20%. This can be understood by considering the Clapeyron equation, which states that the slope of the coexistence curve is inversely proportional to the difference in volume between the phases. If the liquid volume is significantly

mispredicted at the triple point, which is the case for SRK and PR, the melting curve will also be incorrect.

The relative deviations in melting pressures in Figure 1a are plotted up to 300 K, corresponding to high pressures exceeding 500 MPa. In the SI, we plot the relative deviation in the melting temperature as a function of pressure, showing that all models (also PR and SRK) are accurate to 1% in the range of pressures seen in Table 2.

Figure 1b shows that including experimental data below the triple point in the regression of the α -function of tc-PR, as described in Section 3.2, reduces the error in the sublimation curve from about 15% at 140 K to within the experimental uncertainty. We shall see in Section 4 that it also significantly improves the SLE and SVLE for the mixtures.

We find that both of the multiparameter EoS considered, EoS-CG and GERG-2008, give melting and sublimation curves that

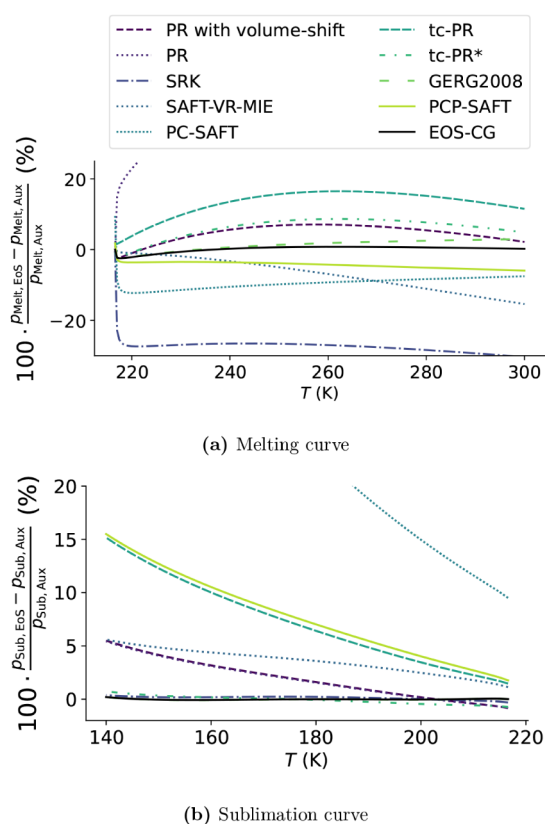


Figure 1. Relative deviation between the melting (a) and sublimation (b) pressures of pure CO₂ predicted by different EoS and the auxiliary functions provided by Trusler.¹⁴ tc-PR* means that the α -function has been modified as elaborated in Section 3.2. The pure-phase EoS for EoS-CG is the one by Span and Wagner;⁷⁷ for GERG-2008, the pure-phase EoS is the one by Klimeck.⁷⁸

are nearly within the experimental uncertainty. EoS founded in SAFT tend to underestimate the melting line, and the sublimation curve appears to be significantly overpredicted.

4.1.2. SVE Predictions. Prediction of solid–vapor equilibria (SVE) with different EoS is compared to experimental data in the first part of Table 3. All EoS, with the exception of PC-SAFT, yield values below 0.8%. When comparing to available experimental data for the CO₂/CH₄ system, we find that the EoS gives similar predictions. More figures that demonstrate this have been included in SI. Figure 2 compares predictions to experimental data for the CO₂/N₂ system.^{54,55} For this mixture, PR gives the best agreement with the experimental data at pressures exceeding 5 MPa. Both EoS-CG and tc-PR are close to the experimental data, with small deviations at higher pressures, while SAFT-VR-Mie and PCP-SAFT exhibit the largest deviations. PC-SAFT also exhibits large deviations that are similar to those of the SAFT-VR-Mie (not shown).

4.1.3. SLE Predictions. Prediction of Solid–Liquid Equilibria (SLE) is compared to experimental data in the midpart of Table 3. The table shows that the choice of fluid EoS has a large impact on the accuracy of the SLE predictions. Rather surprisingly, PR gives the overall best agreement with experimental data.

A comparison of the difference between the predicted and experimental solid formation temperatures for CO₂-mixtures with five components relevant for CCS: CH₄, Ar, H₂S, N₂, and O₂ is compared for the two most accurate choices of EoS in Figure 3. Similar plots for the other EoS considered in this work can be found in SI.

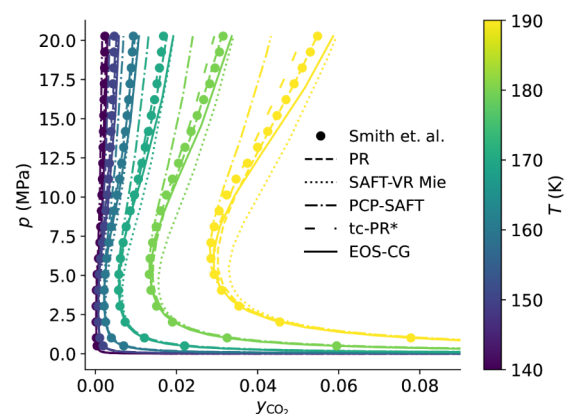


Figure 2. Solid–vapor equilibria of a carbon dioxide and nitrogen mixture for six isotherms, predicted with different EoS for the fluid phase and compared to measurements from refs 54 and 55.

For the CO₂/CH₄ mixture, EoS-CG gives notably higher accuracy than PR, in particular at low CO₂-concentrations. Also for the CO₂–Ar system, EoS-CG reproduces the experimental SLE data near the triple point, nearly within the experimental uncertainty.

PR gives a larger spread in predictions than EoS-CG for the CO₂/CH₄ mixture but gives more accurate predictions for CO₂ mixtures with oxygen and nitrogen. These mixtures were measured at very low temperatures ranging from around 80–140 K with mole fractions in the parts per million (PPM). For SLE predictions for the CO₂/O₂ system at lower temperatures, EoS-CG is unreliable, with deviations exceeding 10 K. This is also evident from Figure 4, which compares SLE predictions to experimental data from two different sources.^{58,59} A potential for improvement is to revise the binary departure function of the CO₂/O₂ mixture of EoS-CG with new experimental data,⁷⁹ which would likely improve SLE predictions.

We find that including experimental data below the triple point in the regression of the α -function of cubic EoS, as described in Section 3.2, reduces the MAPD in predicting SLE in mixtures from 2.4% (original) to 1.6% (modified), even if only experimental data for pure CO₂ were used in the regression. Interestingly, standard PR gives an even lower MAPD of 1.4% for SLE predictions without regressing any parameters to experimental data below the triple point. This suggests that it might be possible to further improve the overall performance of tc-PR by including another parameter in the temperature-dependent α -function.

Table 3 shows that PC-SAFT and SAFT-VR-Mie give inaccurate predictions of SLE with solid CO₂. A possible explanation for this is that their representation of pure CO₂ is not sufficiently accurate. PCP-SAFT performs significantly better, with a MAPD of 2.4% for SLE predictions, compared to 6.8% for PC-SAFT and 6.3% for SAFT-VR-Mie. This demonstrates the importance of incorporating quadrupole effects into the SAFT framework to represent CO₂ accurately.

4.1.4. SVLE Predictions. Prediction of Solid–Vapor–Liquid Equilibria (SVLE) is compared to experimental data in the last part of Table 3. SVLE curves make it possible to assess solid formation predictions in the critical region, where the deviations between the predictions and experimental data are usually larger. The SVLE locus of the CO₂/CH₄ SVLE mixture is shown in Figure 5. The largest spread between models is in the critical region at the top of the curve, where EoS-CG is closest to the

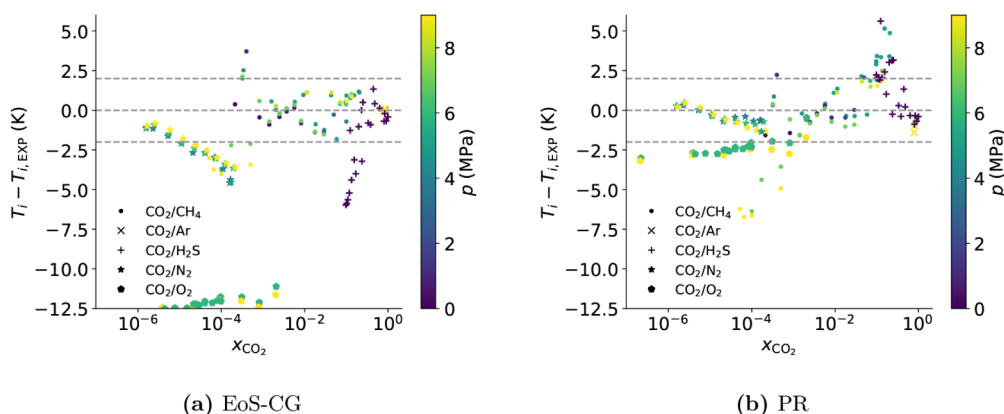


Figure 3. Difference between the predicted T_i and experimental $T_{i,EXP}$ temperatures from SLE in several mixtures for EoS-CG (a) and PR (b). See Table 2 for details of the experimental data.

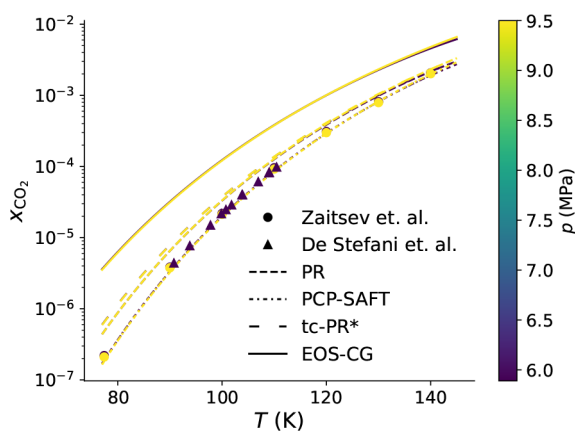


Figure 4. SLE curve for a CO_2/O_2 mixture at different isobars compared with experimental data;^{58,59} see Table 2 for details.

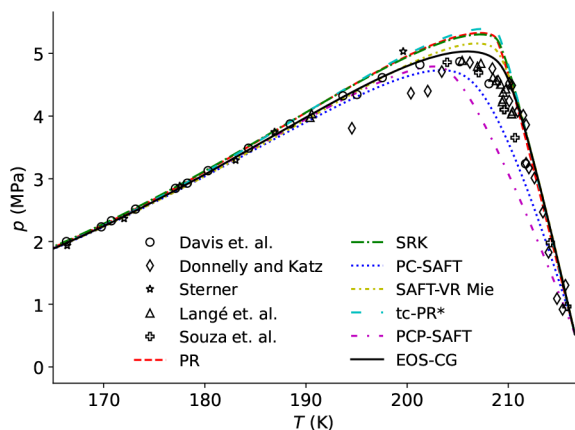


Figure 5. Predicted SVLE locus in a methane– CO_2 mixture compared with SVLE locus data;^{42–44,65,66} see Table 2 for details.

experimental data. The cubic EoS overpredicts, while the PC-SAFT variants underpredict the experimental pressures in the critical region. The behavior sufficiently close to the critical point is nonanalytic. Therefore, none of the EoS used is expected to provide a reasonable description in this region.

The SVLE envelopes of the CO_2/N_2 and CO_2/H_2 mixtures are shown in Figure 6. Here, PR lies closest to the experimental data for CO_2/N_2 , and both PR and EoS-CG perform well for CO_2/H_2 . The SVLE envelopes of mixtures that are highly

relevant for natural gas applications, i.e., CO_2 -mixtures with H_2S , ethane, methane, and *n*-butane, are shown in Figure 7. For natural gas applications, PR, EoS-CG, and GERG-2008 perform well, while PC-SAFT and SAFT-VR-Mie exhibit large deviations from the experimental data. Due to better performance in the critical region, EoS-CG performs best for SVLE predictions.

We find that all of the fluid EoS yield MAPDs exceeding 2% when predicting dry ice formation in mixtures when the mole fractions are in the per million range. This is the case for the measurements by Riva et al.⁶³ and is probably linked to a higher uncertainty of the experimental measurements.

By summarizing the MAPD of all CO_2 -mixtures with hydrocarbons from Table 3, we find that EoS-CG performs best with an MAPD below 0.9%, while both PR and GERG-2008 are also excellent choices with a MAPD of 1.1%. If nitrogen, oxygen, or SO_2 are important components in the mixture, as in CCS applications, PR is the best choice.

5. CONCLUSION

We have reviewed available experimental data on vapor and liquid in equilibrium with solid CO_2 for mixtures relevant for carbon capture, transport, and storage (CCS) and natural gas applications. There are many measurements available for the CO_2/CH_4 mixture, but more measurements are needed for binary and multicomponent CO_2 -mixtures with H_2 , O_2 , N_2 , Ar, H_2S , SO_2 , in particular for CO_2 -rich systems.

The conditions where solid CO_2 forms depend on the composition of the mixture. They can be predicted by combining an Equation of State (EoS) for the fluid phases and an EoS for the solid phase with a suitable phase-equilibrium algorithm. We tested a wide selection of model combinations to determine the best choice in order to establish a benchmark choice of models to estimate when dry ice forms from mixtures and to estimate the uncertainty of this model combination.

The two most accurate EoS for solid CO_2 by Trusler¹⁴ and by Jäger and Span¹⁶ were found to give very similar predictions. A wide range of EoS was tested for the fluid phase: three cubic EoS, Soave–Redlich–Kwong (SRK), Peng–Robinson (PR), and tc-PR; three EoS founded in Statistical Associating Fluid Theory (SAFT), PC-SAFT, SAFT-VR-Mie, and PCP-SAFT; and two multiparameter EoS, GERG-2008, and EoS-CG. A key factor in the success of predicting freeze-out is the fluid EoS's extrapolative behavior to low temperatures.

For solid–vapor equilibria, all fluid EoS gave reasonable predictions with Mean Absolute Percentage Deviations

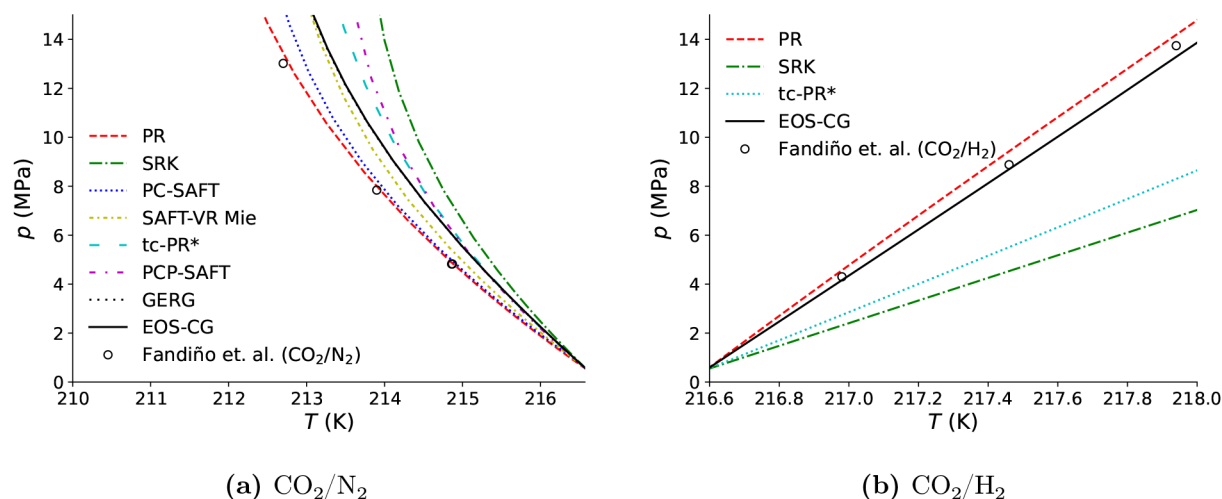


Figure 6. Predicted SVLE locus for the CO_2/N_2 (a) and CO_2/H_2 (b) mixtures compared with experimental data from ref 68.

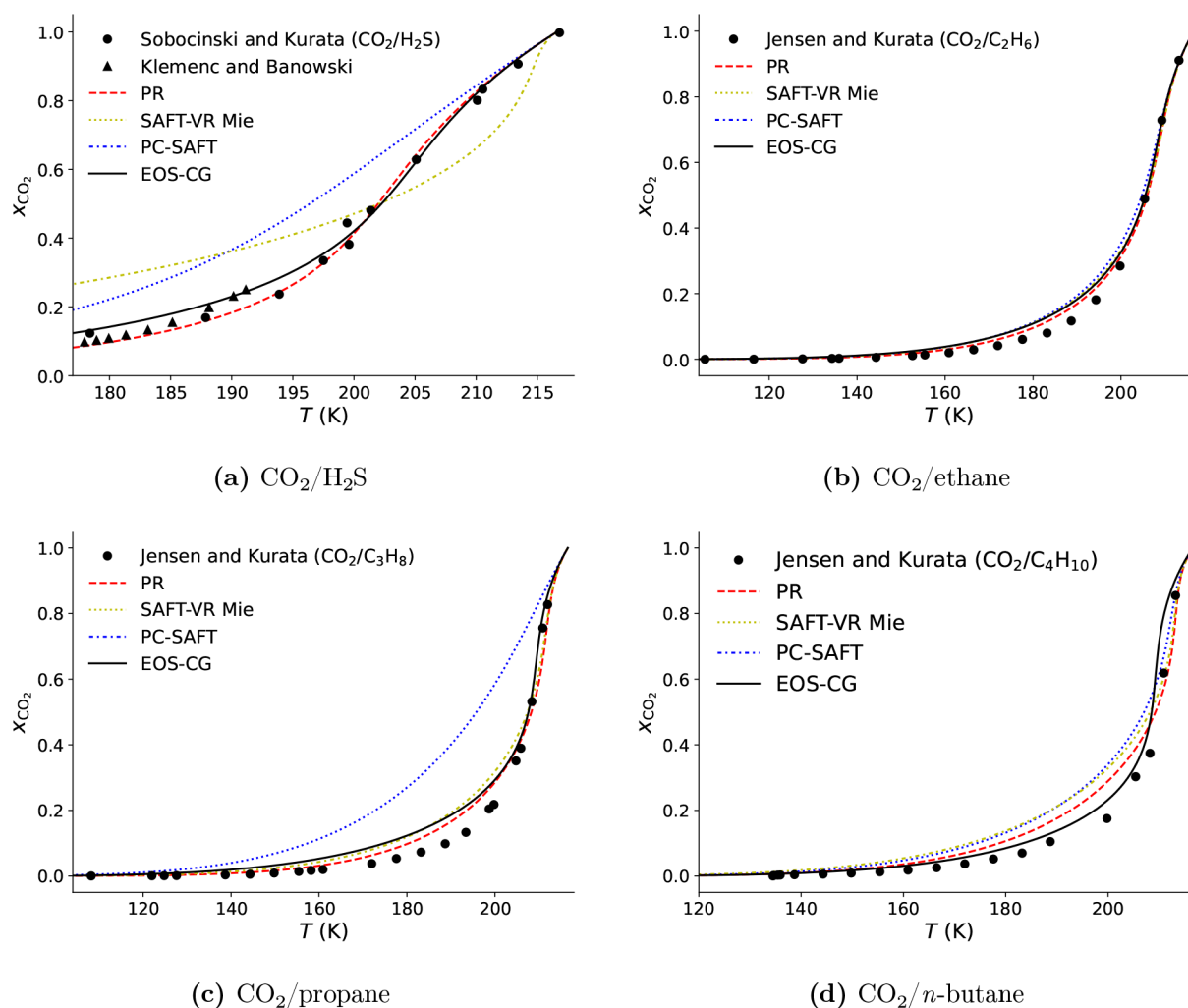


Figure 7. Predicted SVLE locus for binary CO_2 mixtures, H_2S (a), ethane (b), propane (c), and n -butane (d), compared with SVLE-locus data;^{60,61,67} see Table 2. Since GERG-2008 gives very similar results as EoS-CG and SRK and tc-PR give very similar results as PR for these calculations, they are not shown in the figure.

(MAPDs) below 1.1%. For solid–liquid and solid–vapor–liquid equilibria, the choice of fluid EoS had a larger influence on the accuracy.

Overall, standard PR gave the best agreement with all of the experimental data, with an overall MAPD of 1.4%. For natural gas applications where hydrocarbons are the main components, EoS-CG was the most accurate choice with a MAPD of 0.9%.

The largest deviations for EoS-CG were for the CO₂/O₂ mixture, where the binary departure function should be improved.

Since standard PR gave a better agreement with experimental data than tc-PR, the α -function of tc-PR was refitted by considering also experimental data for pure CO₂ below the triple point. This improved the MAPD for solid–liquid equilibria from 2.4% (original) to 1.6% (modified), even if only experimental data for pure CO₂ were used in the regression. It also reduced the error in the predicted sublimation curve of pure CO₂ from about 15% at 140 K to within experimental uncertainty. Provided that an accurate EoS for the solid phase is available, including sublimation pressure data in the regression of the fluid EoS represents an avenue that can be explored to improve solid–fluid equilibrium predictions also for other components in the future.

PC-SAFT and SAFT-VR-Mie give significantly less accurate predictions than the other EoS, probably due to an inaccurate representation of pure CO₂. Including the quadrupole moment of CO₂, as done in PCP-SAFT, improves the agreement with the experiments.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.5c04028>.

Information about the binary interaction parameters used, a comparison of the performance of tc-PR EoS before and after modifying the α -function, and additional comparisons and figures with solid–fluid equilibrium (SFE) predictions (PDF)

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Notes

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